Assignment 2 -final

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Politecnico di Torino - Master of Science in Communications Engineering

Communication Systems
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Assignment 2

- Exercise 1 - Hard Correction vs. Soft Correction pt. 8+2
- Exercise 2 - Linear Feedback Shift Registers and m-sequences pt. 8
- Exercise 3 - Sync Marker detection pt. 6+2
- Question 1 - Extension, Shortening, Puncturing pt. 1.5
- Question 2 - Autocorrelation property of m-sequences pt. 1.5
- Question 3 - Autocorrelation and fft pt. 1

Exercise 1 - Hard Correction vs. Soft Correction (pt. 8)

Fix $k = 4$ and $n = 8$.
Consider this generator matrix (equal to matrix G_2 of Assignment 1):

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Hard Correction vs. Soft Correction (pt. 12 (?))

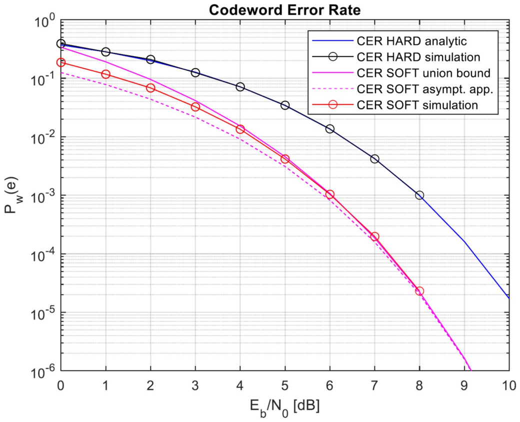
1. Consider $E_b/N_0 = 0 : 10$ dB. Compute and plot the Hard Correction Codeword Error Rate analytic curve.
2. Build the LUT corresponding to all the error vectors with $w_H\left(\underline{e}\right) \leq t$.
3. Simulate and plot the Codeword Error Rate obtained by applying Syndrome Decoding with this LUT. Use $E_b/N_0 = 0 : 7$ dB (or less if your PC is too slow).

Hard Correction vs. Soft Correction

4. Consider $E_b/N_0 = 0 : 10$ dB. Compute and plot the Soft Correction Codeword Error Rate union bound and asymptotic approximation.
5. Simulate and plot the Codeword Error Rate obtained by applying Soft Correction with the minimum Euclidean distance (or maximum correlation) decoding rule. Use $E_b/N_0 = 0 : 7$ dB (or less if your PC is too slow).

Hard Correction vs. Soft Correction

6. Comment the results.

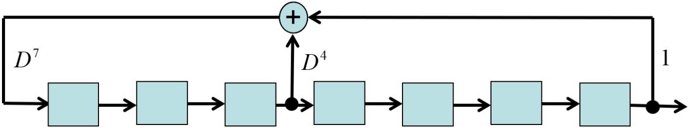

Codeword Error Rate
CER HARD analytic
CER HARD simulation
CER SOFT union bound
CER SOFT asympt. app.
CER SOFT simulation
$P_w(e)$
E_b/N_0 [dB]
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
1
2
3
4
5
6
7
8
9
10

Second part = Syndrome Decoding with the full LUT (pt. 2)

- Generate the entire LUT with size 2^r, by adding the missing syndromes corresponding to error vectors with $w(\underline{e}) > t$.
- Simulate and plot (add the curve to the previous figure) the Codeword Error Rate obtained by applying Syndrome Decoding with this full LUT. Use $E_b/N_0 = 0:7$ dB (or less if your PC is too slow).
- Comment the results.

Exercise 2 - LFSR m-sequences (pt. 8)

The Primary Synchronization Signal (PSS) of 5G New Radio uses an m-sequence generated by this Linear Feedback Shift Register (LFSR):



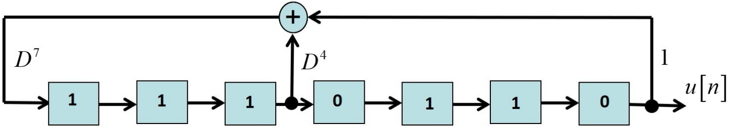
The LFSR is characterized by;

- m=7 cells
- polynomial description $D^7 + D^4 + 1$

(The association between the feedback links and the polynomial coefficients is not unique, the reverse order is used too. The proposed one is the association used by Matlab in its functions.)

Starting seed

The starting seed is 1110110:



The polynomial is primitive then the LFSR generates an m-sequence with

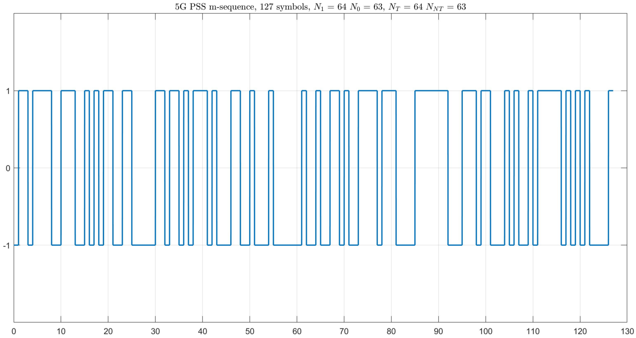
- period $N = 2^m - 1 = 127$ bits
- first bits = 0110111100...

Denote the binary sequence by $v(n)$ and the corresponding bipolar sequence ($0 \rightarrow -1$, $1 \rightarrow +1$) by $v'(n)$, for $0 \leq n \leq N - 1$.
When needed, consider them as the principal periods of periodic sequences.

0/1 and transition properties

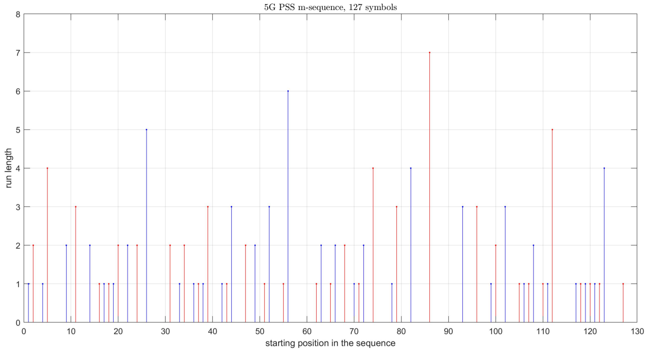
1. Generate and plot the 127-symbol bipolar sequence $v'(n)$.
2. Write a table with
   - N_1 and N_0 (number of bits equal to 1 or 0 in $v(n)$)
   - N_T and N_{NT} (number of transitions and no-transitions in $v(n)$)
3. Check if the N_1/N_0 and N_T/N_{NT} properties of m-sequences are verified.

Bipolar sequence



Runs

4. Write a table with the values of $NR_0(i)$ and $NR_1(i)$ (number runs of i consecutive 0 or 1 symbols in $v(n)$)
5. Plot the run lengths vs. their starting points in the sequence (use different colors for 0 and 1 runs)
6. Compare the values of $NR(0)$ and $NR(1)$ against the run properties of an ideal binary random sequence and discuss the results.


5G PSS m-sequence, 127 symbols
run length
starting position in the sequence
0
1
2
3
4
5
6
7
8
0
10
20
30
40
50
60
70
80
90
100
110
120
130

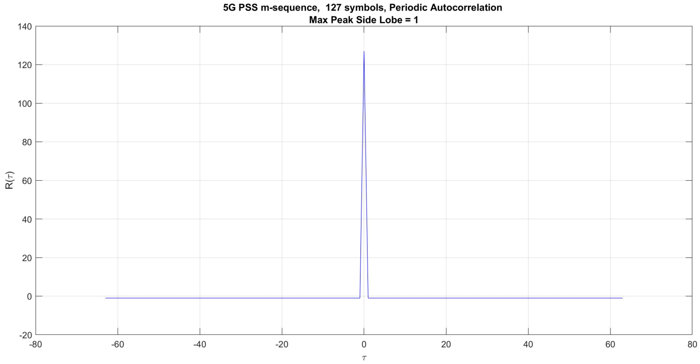
Autocorrelation

7. Compute and plot the periodic autocorrelation function

$$R(\tau) = \sum_{n=0}^{N-1} b(n)\, b(n-\tau) \qquad -(N-1) \leq \tau \leq N-1$$

8. Verify the property:
   - $R(\tau) = N$ for $\tau = 0$
   - $R(\tau) = -1$ for $\tau \neq 0$

Autocorrelation plot



Truncated m-sequences

Cancel the last $i = 10$ bits of the m-sequence

9. Plot the autocorrelation function of the truncated sequence of $N = 117$ symbols.
10. Compute the Maximum Peak Side Lobe (MPSL)

$$MPSL = max_{\tau \neq 0} \left| R\left(\tau\right) \right|$$

11. Comment the result.
12. Repeat with another starting seed.
13. Comment the result.

Matlab

LFSR

```
m=7; % number of cells
Nb=2∧m-1; % period
pnSequence = comm.PNSequence('Polynomial',[7 4 0], ...
'SamplesPerFrame',Nb,'InitialConditions',[1 1 1 0 1 1 0]);
x1 = pnSequence()';
```

Matlab

Autocorrelation

```
x1b=2*x1-1; % bipolar version 0 → -1 1 → +1
R=ifft(fft(x1b).*conj(fft(x1b))); % non-normalized periodic autocorr.
```

Exercise 3 - Sync Marker detection (pt. 6)

- Fix the sync marker length $N = 16$.
- Generate a random binary pattern $\underline{p}$ with length N.
- Map it into a bipolar sequence $\underline{p'} = 2\underline{p} - 1$.
- Fix SNR $= E_s/\sigma^2 = -10$ dB.
- Fix the number of simulations (e.g. $N_s = 100,000$).
- Fix the minimum number of observed events (e.g. $N_e = 20$);

Simulations

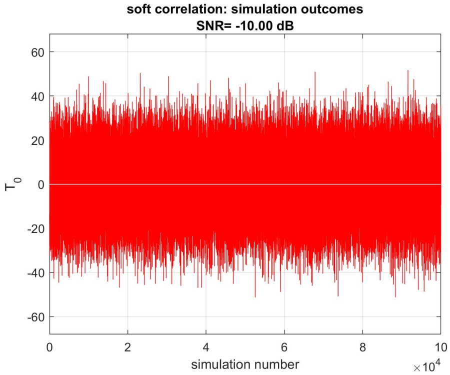
For each simulation point:

- Generate a noise pattern $\underline{n} = randn(1, N) * \sigma$.
- Consider $\underline{r}_0 = \underline{n}$ (received signal under H_0) and $\underline{r}_1 = \underline{p'} + \underline{n}$ (received signal under H_1).
- Compute the statistical tests $T_0 = \sum_{i=1}^{N} r_{0,i} p'_i$ and $T_1 = \sum_{i=1}^{N} r_{1,i} p'_i$

Results

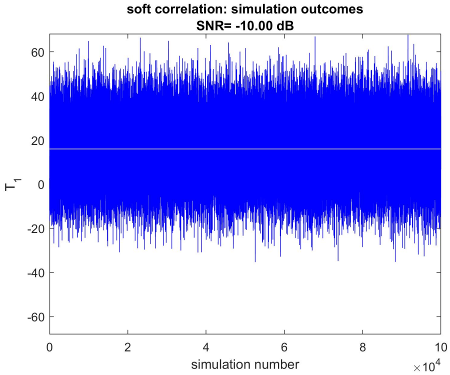
1. Plot T_0 and T_1
2. Verify the correctness of their mean values.

T_0 and T_1



soft correlation: simulation outcomes
SNR= -10.00 dB
60
40
20
0
-20
-40
-60
T_0
0
2
4
6
8
10
$\times 10^4$
simulation number

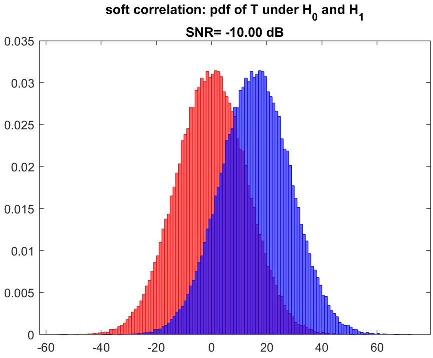
T_0 and T_1


soft correlation: simulation outcomes
SNR= -10.00 dB
60
40
20
0
-20
-40
-60
T_1
0
2
4
6
8
10
simulation number
$\times 10^4$

Results

3. Use *histogram* to plot the pdfs of T_0 and T_1
4. Comment the results

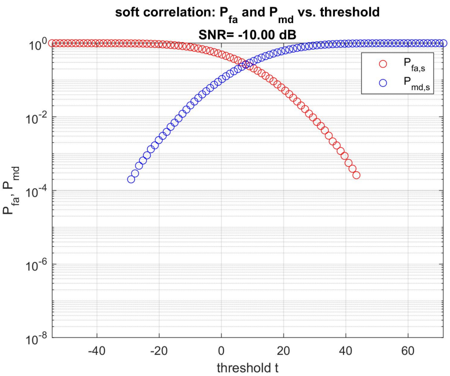
Probability density functions



Results

5. Compute and plot P_{fa} and P_{md}
6. Comment the results

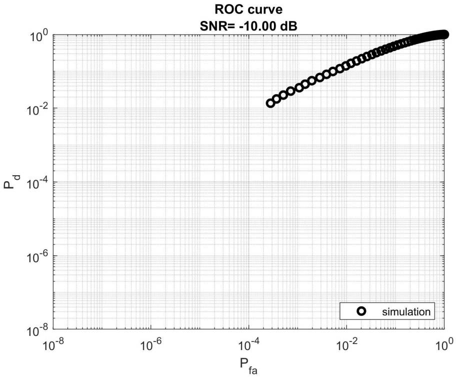
False alarm and missed detection probabilities: simulation



Receiver Operating Characteristic

7. Plot the ROC curve $P_d = (1 - P_{md})$ vs. P_{fa}

ROC: simulation



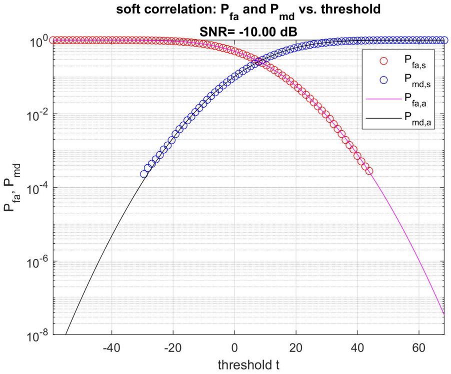
Receiver Operating Characteristic

8. Repeat the experiment for different values of SNR. Comment the results.
9. Repeat the experiment for different values of N. Comment the results.

Second Part = Analytic curves (pt. 2)

1. Compute and add the analytic curves of P_{fa} and P_{md}

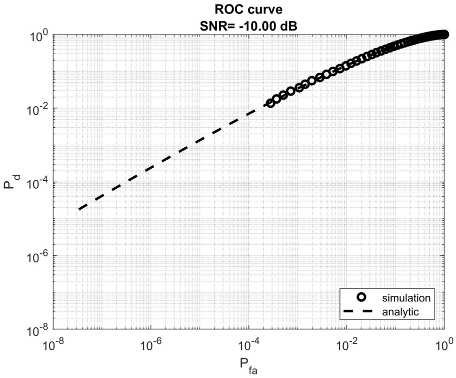
P_{fa} and P_{md}: simulated vs. analytic curves



ROC analytic curve

2. Add the analytic ROC curve

ROC: simulation vs. analytic curves

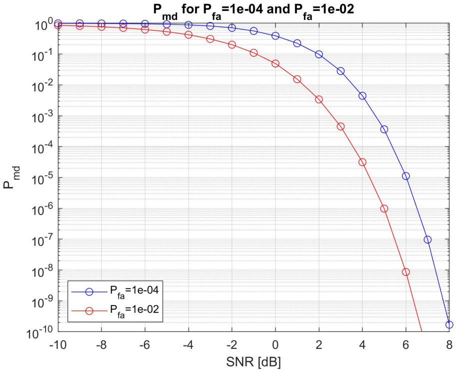


P_{md} vs. SNR

Consider $N = 16$ and $SNR_{dB} = -10 : 8$ and compute analytically the value of P_{md} corresponding to $P_{fa} = 1e - 2$ and $P_{fa} = 1e - 4$:

3. Plot the curves of P_{md} for $P_{fa} = 1e - 2$ and $P_{fa} = 1e - 4$ vs. SNR [dB]
4. Comment the results

P_{md} vs. SNR



P_md for P_fa=1e-04 and P_fa=1e-02
P_md
10^0
10^-1
10^-2
10^-3
10^-4
10^-5
10^-6
10^-7
10^-8
10^-9
10^-10
-10
-8
-6
-4
-2
0
2
4
6
8
SNR [dB]
P_fa=1e-04
P_fa=1e-02

Question 1: extension, puncturing, shortening (pt. 1.5)

Given a code $C(n,k)$ with minimum distance $d_{min}(C) = d$ we can obtained three codes by the following operations:

- Extension: the code $C_1(n_1, k_1)$ is obtained by adding an extra parity bit, such that all the codewords have an even weight. Then $k_1 = k$ and $n_1 = n + 1$.
- Puncturing: the code $C_2(n_2, k_2)$ is obtained by deleting a parity bit, that is not transmitted. Then $k_2 = k$ and $n_2 = n - 1$.
- Shortening: the code $C_3(n_3, k_3)$ is obtained by considering only the codewords of C with a zero in the first position and by deleting this first bit. Then $k_3 = k - 1$ and $n_3 = n - 1$.

Which is the relationship between:

- $d_{min}(C)$ and $d_{min}(C_1)$
- $d_{min}(C)$ and $d_{min}(C_2)$
- $d_{min}(C)$ and $d_{min}(C_3)$?

Question 2: autocorrelation property of m-sequences (pt. 1.5)

Prove that for an entire m-sequence, this property holds:

- $R(\tau) = N$ for $\tau = 0$
- $R(\tau) = -1$ for $\tau \neq 0$

[Hint: consider a code with $k = m$ (number of cell of the LFSR)]

Question 3: autocorrelation and fft (pt. 1)

Prove that the autocorrelation can be computed by the fft.

$$R = ifft(fft(x1b). * conj(fft(x1b)));$$

Deadline for +2 bonus: 30/11/2022 11 PM